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Practical No.-02

Aim: -Working of Digital Differential Analyzer Algorithm.

Theory: - DDA (Digital Differential Analyzer) is a line drawing algorithm used in computer graphics to generate a line segment between two specified endpoints. It is a simple and efficient algorithm that works by using the incremental difference between the x-coordinates and y-coordinates of the two endpoints to plot the line.

DDA algorithm is relatively easy to implement and is computationally efficient, making it suitable for real-time applications. However, it has some limitations, such as the inability to handle vertical lines and the need for floating-point arithmetic, which can be slow on some systems. Nonetheless, it remains a popular choice for generating lines in computer graphics.

- It is a simple and easy-to-implement algorithm.
- It avoids using multiple operations which have high time complexities.
- It is faster than the direct use of the line equation because it does not use any floating-point multiplication and it calculates points on the line.

Algorithm: -

The steps involved in DDA line generation algorithm are:

1. Input the two endpoints of the line segment, (x1, y1) and (x2, y2).
2. Calculate the difference between the x-coordinates and y-coordinates of the endpoints as dx and dy respectively.
3. Calculate the slope of the line as m = dy/dx.
4. Set the initial point of the line as (x1, y1).
5. Loop through the x-coordinates of the line, incrementing by one each time, and calculate the corresponding y-coordinate using the equation y = y1 + m (x – x1).
6. Plot the pixel at the calculated (x, y) coordinate.
7. Repeat steps 5 and 6 until the endpoint (x2, y2) is reached.

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Program Code: -

```
from matplotlib import pyplot as plt

# DDA Function for line generation


def DDA(x0, y0, x1, y1):

  # find absolute differences
  dx = abs(x0 - x1)
  dy = abs(y0 - y1)

  # find maximum difference
  steps = max(dx, dy)

  # calculate the increment in x and y
  xinc = dx/steps
  yinc = dy/steps

# start with 1st point
  x = float(x0)
  y = float(y0)

  # make a list for coordinates
  x_coorinates = []
  y_coorinates = []

  for i in range(steps):
    # append the x,y coordinates in respective list
    x_coorinates.append(x)
    y_coorinates.append(y)

    # increment the values
    x = x + xinc
    y = y + yinc
```

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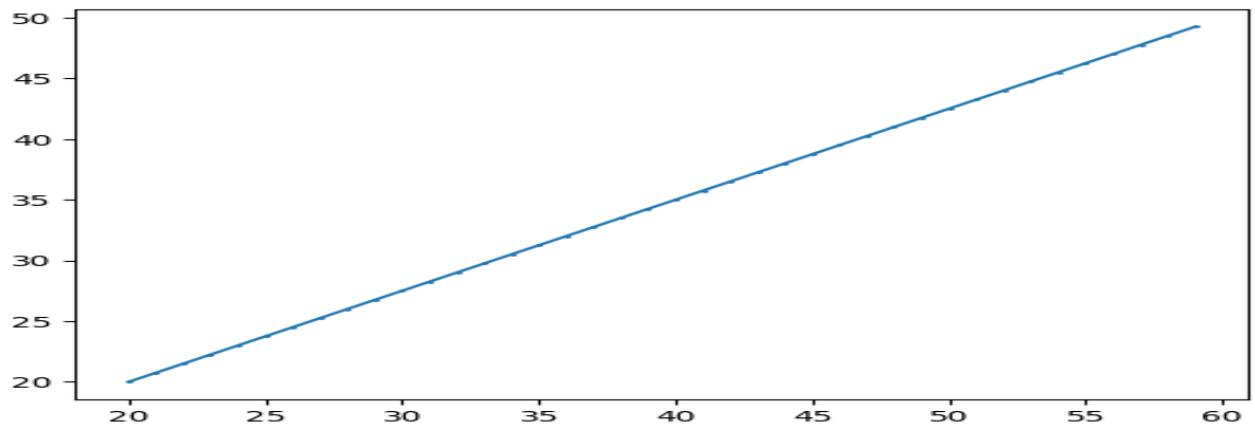
```
  # plot the line with coordinates list
  plt.plot(x_coorinates, y_coorinates, marker="o",
      markersize=1, markerfacecolor="green")
  plt.show()

# Driver code
if __name__ == "__main__":

  # coordinates of 1st point
  x0, y0 = 20, 20

  # coordinates of 2nd point
  x1, y1 = 60, 50

  # Function call
  DDA(x0, y0, x1, y1)

  # This code is contributed by 111arpit1
```

Output: -



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Conclusion: - Thus, I have implemented Working of Digital Differential Analyzer Algorithm successfully.

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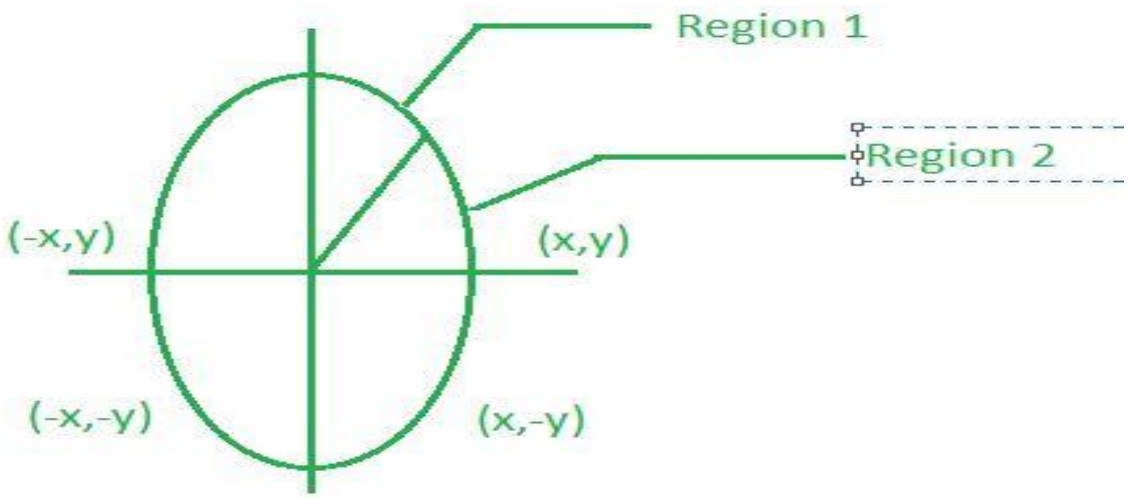
Practical No.-03

Aim: - Study and implementation of Ellipse Generation Algorithm.

Theory: -

Mid-point Ellipse algorithm is used to draw an ellipse in computer graphics.



Midpoint ellipse algorithm plots(finds) points of an ellipse on the first quadrant by dividing the quadrant into two regions.
Each point(x, y) is then projected into other three quadrants (-x, y), (x, -y), (-x, -y) i.e. it uses 4-way symmetry.

Function of ellipse:

$f_{ellipse}(x, y)=r_y^2x^2+r_x^2y^2-r_x^2r_y^2$
$f_{ellipse}(x, y)<0$ then (x, y) is inside the ellipse.
$f_{ellipse}(x, y)>0$ then (x, y) is outside the ellipse.
$f_{ellipse}(x, y)=0$ then (x, y) is on the ellipse.

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Algorithm: -

1. Take input radius along x axis and y axis and obtain center of ellipse.
2. Initially, we assume ellipse to be centered at origin and the first point as : $(x, y_0)= (0, r_y)$.
3. Obtain the initial decision parameter for region 1 as: $p1_0=r_y^2+1/4r_x^2-r_x{}^2r_y$
4. For every x_k position in region 1 :
   If $p1_k<0$ then the next point along the is (x_{k+1}, y_k) and $p1_{k+1}=p1_k+2r_y^2x_{k+1}+r_y^2$
   Else, the next point is (x_{k+1}, y_{k-1})
   And $p1_{k+1}=p1_k+2r_y^2x_{k+1} – 2r_x^2y_{k+1}+r_y^2$
5. Obtain the initial value in region 2 using the last point (x_0, y_0) of region 1 as:
   $p2_0=r_y^2(x_0+1/2)^2+r_x^2 (y_0-1)^2-r_x^2r_y^2$
6. At each y_k in region 2 starting at k =0 perform the following task.
   If $p2_k>0$ the next point is (x_k, y_{k-1}) and $p2_{k+1}=p2_k-2r_x^2y_{k+1}+r_x^2$
7. Else, the next point is (x_{k+1}, y_{k-1}) and $p2_{k+1}=p2_k+2r_y^2x_{k+1} -2r_x^2y_{k+1}+r_x^2$
8. Now obtain the symmetric points in the three quadrants and plot the coordinate value as: x=x+xc, y=y+yc
9. Repeat the steps for region 1 until $2r_y^2x$>$=2r_x^2y$

Program Code: -

```
import matplotlib.pyplot as plt

def draw_ellipse(a, b):

    # Initialize variables

    x = 0

    y = b

    a2 = a * a

    b2 = b * b
```

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```
# Decision parameter for Region 1
p1 = b2 - (a2 * b) + (0.25 * a2)
dx = 2 * b2 * x
dy = 2 * a2 * y
points = []
# Region 1: Slope > 1
while dx < dy:
    points.append((x, y))
    if p1 < 0:
        x += 1
        dx += 2 * b2
        p1 += dx + b2
    else:
        x += 1
        y -= 1
        dx += 2 * b2
        dy -= 2 * a2
```

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```
        p1 += dx - dy + b2
# Decision parameter for Region 2
p2 = (b2 * (x + 0.5) ** 2) + (a2 * (y - 1) ** 2) - (a2 * b2)
# Region 2: Slope <= 1
while y >= 0:
    points.append((x, y))
    if p2 > 0:
        y -= 1
        dy -= 2 * a2
        p2 += a2 - dy
    else:
        y -= 1
        x += 1
        dx += 2 * b2
        dy -= 2 * a2
        p2 += dx - dy + a2
return points
```

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```
def plot_ellipse(a, b):
    # Get points
    ellipse_points = draw_ellipse(a, b)
    # Use symmetry to plot all quadrants
    x_coords = []
    y_coords = []
    for x, y in ellipse_points:
        x_coords.extend([x, -x, x, -x])
        y_coords.extend([y, y, -y, -y])
    # Plot the ellipse
    plt.figure(figsize=(6, 6))
    plt.plot(x_coords, y_coords, 'bo', markersize=1)
    plt.title(f"Ellipse with a={a}, b={b}")
    plt.xlabel("X-axis")
    plt.ylabel("Y-axis")
    plt.axhline(0, color='black', linewidth=0.5)
    plt.axvline(0, color='black', linewidth=0.5)
```

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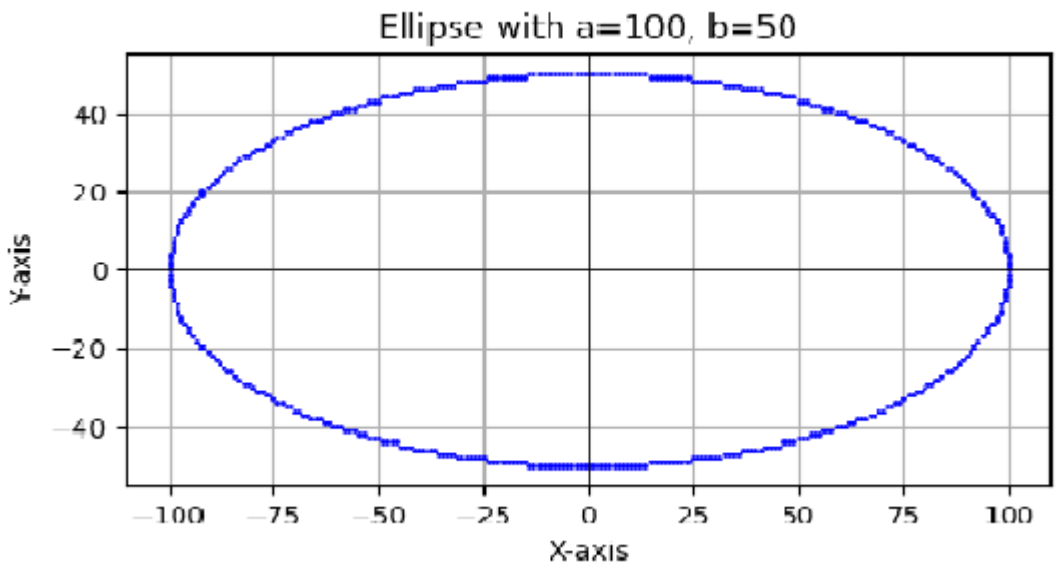
```
    plt.grid()

    plt.gca().set_aspect('equal', adjustable='box')

    plt.show()

# Example usage

plot_ellipse(100, 50)
```

Output: -



Conclusion: - Thus, I have Study and implemented of Ellipse Generation Algorithm successfully.

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Practical No.-04

Aim: - Implementation and use of Creating two dimensional objects.

Theory: -

Two-dimensional objects form the basis of computer graphics. They can be represented mathematically and drawn on a screen using algorithms or libraries. Common 2D shapes include:

1. **Lines:** Defined by two endpoints.
2. **Rectangles:** Defined by width, height, and a starting point.
3. **Circles:** Defined by a center and radius.
4. **Polygons:** Defined by a series of vertices.

In Python, libraries such as **Matplotlib** or **Pygame** are widely used for creating and rendering 2D graphics.

Algorithm: -

• Initialization**:**

- Set up the graphics environment using a Python library (e.g., Matplotlib or Pygame).

• Object Creation:

- **Line:** Plot a straight line between two points.
- **Rectangle:** Plot a rectangle using four points or by specifying width and height.
- **Circle:** Calculate points on the circumference using the circle equation (x−h)2+(y−k)2=r2(x - h)^2 + (y - k)^2 = r^2(x−h)2+(y−k)2=r2.
- **Polygon:** Connect vertices in sequence to create a closed shape.

• Rendering:

- Use library functions to display the created objects on a 2D coordinate system.

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Program Code: -

```
import matplotlib.pyplot as plt
import numpy as np

def draw_line(x1, y1, x2, y2):
    plt.plot([x1, x2], [y1, y2], label="Line")

def draw_rectangle(x, y, width, height):
    rectangle = plt.Rectangle((x, y), width, height, fill=False,
label="Rectangle")
    plt.gca().add_patch(rectangle)

def draw_circle(x, y, radius):
    circle = plt.Circle((x, y), radius, fill=False, label="Circle")
    plt.gca().add_patch(circle)

def draw_polygon(vertices):
    polygon = plt.Polygon(vertices, fill=False, closed=True, label="Polygon")
    plt.gca().add_patch(polygon)

def create_2d_objects():
```

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```
# Line
draw_line(0, 0, 5, 5)

# Rectangle
draw_rectangle(1, 1, 4, 2)

# Circle
draw_circle(3, 3, 2)

# Polygon
vertices = [(2, 5), (3, 8), (5, 7), (4, 4)]
draw_polygon(vertices)

# Display settings
plt.gca().set_aspect('equal', adjustable='box')
plt.axhline(0, color='black', linewidth=0.5)
plt.axvline(0, color='black', linewidth=0.5)
plt.grid(color='gray', linestyle='--', linewidth=0.5)
plt.legend(loc="upper left")
plt.title("2D Objects in Python")
```

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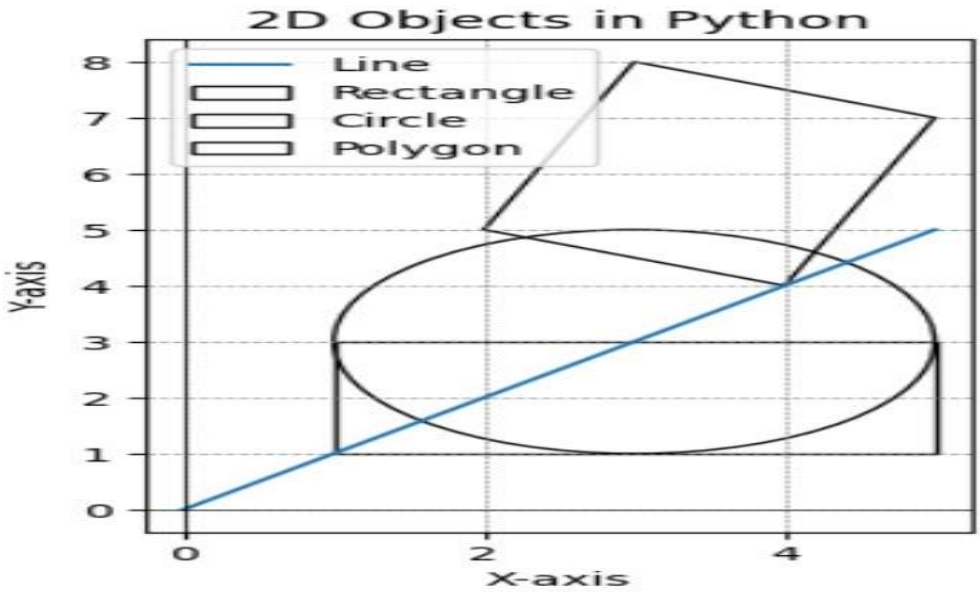
```
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.show()

# Run the function to create and display 2D objects
create_2d_objects()
```

Output:-



Conclusion: - Thus, I have Implementation and use of Creating two dimensional objects successfully.

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Practical No.-05

Aim: - Study and implementation of Two-Dimensional Transformations.

Theory: -

Two-dimensional transformations are mathematical operations applied to objects to modify their position, size, orientation, or shape in a 2D plane.

Types of Transformations:

1. **Translation:** Moves an object from one position to another in the 2D plane.
2. **Scaling:** Changes the size of an object.
3. **Rotation:** Rotates an object about the origin by an angle θ\
4. **Reflection:** Flips an object over a line (e.g., x-axis, y-axis, or any arbitrary line).
5. **Shearing:** Distorts an object by shifting its points along an axis.

Algorithm: -

• Input**:**

- Define the object using its vertices (e.g., points for a polygon).
- Specify the transformation parameters (e.g., translation distances, scaling factors, rotation angle).

• Matrix Representation**:** Use matrix operations for transformations. The general transformation matrix for 2D is:

• Apply Transformation:

- Multiply the transformation matrix with the object's coordinates.

• Display**:**

- Plot the original and transformed object for visualization.

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Program Code: -

```
import numpy as np

import matplotlib.pyplot as plt

def plot_polygon(vertices, color, label):

    # Close the polygon for display

    vertices = np.vstack([vertices, vertices[0]])

    plt.plot(vertices[:, 0], vertices[:, 1], color=color, label=label)

def translate(vertices, tx, ty):

    translation_matrix = np.array([

        [1, 0, tx],

        [0, 1, ty],

        [0, 0, 1]

    ])

    vertices_homogeneous = np.hstack([vertices, np.ones((vertices.shape[0], 1))])

    transformed_vertices = vertices_homogeneous @ translation_matrix.T

    return transformed_vertices[:, :2]


def scale(vertices, sx, sy):
```

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```
    scaling_matrix = np.array([
        [sx, 0, 0],
        [0, sy, 0],
        [0, 0, 1]
    ])
    vertices_homogeneous = np.hstack([vertices, np.ones((vertices.shape[0], 1))])
    transformed_vertices = vertices_homogeneous @ scaling_matrix.T
    return transformed_vertices[:, :2]
def rotate(vertices, angle):
    rad = np.radians(angle)
    rotation_matrix = np.array([
        [np.cos(rad), -np.sin(rad), 0],
        [np.sin(rad), np.cos(rad), 0],
        [0, 0, 1]
    ])
    vertices_homogeneous = np.hstack([vertices, np.ones((vertices.shape[0], 1))])
    transformed_vertices = vertices_homogeneous @ rotation_matrix.T
    return transformed_vertices[:, :2]
def reflect(vertices, axis):
```

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```
    if axis == "x":
        reflection_matrix = np.array([
            [1, 0, 0],
            [0, -1, 0],
            [0, 0, 1]
        ])
    elif axis == "y":
        reflection_matrix = np.array([
            [-1, 0, 0],
            [0, 1, 0],
            [0, 0, 1]
        ])
    else:
        raise ValueError("Invalid axis for reflection. Use 'x' or 'y'.")
    vertices_homogeneous = np.hstack([vertices, np.ones((vertices.shape[0], 1))])
    transformed_vertices = vertices_homogeneous @ reflection_matrix.T
    return transformed_vertices[:, :2]
# Example usage
vertices = np.array([[1, 1], [3, 1], [2, 4]])  # Triangle vertices
```

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```
# Plot original object

plot_polygon(vertices, "blue", "Original")

# Apply transformations

translated = translate(vertices, 2, 3)

scaled = scale(vertices, 2, 2)

rotated = rotate(vertices, 45)

reflected = reflect(vertices, "x")

# Plot transformed objects

plot_polygon(translated, "green", "Translated")

plot_polygon(scaled, "red", "Scaled")

plot_polygon(rotated, "orange", "Rotated")

plot_polygon(reflected, "purple", "Reflected")

# Display

plt.axhline(0, color='black', linewidth=0.5)

plt.axvline(0, color='black', linewidth=0.5)

plt.legend(loc="best")

plt.grid(color='gray', linestyle='--', linewidth=0.5)

plt.title("2D Transformations")

plt.xlabel("X-axis")
```

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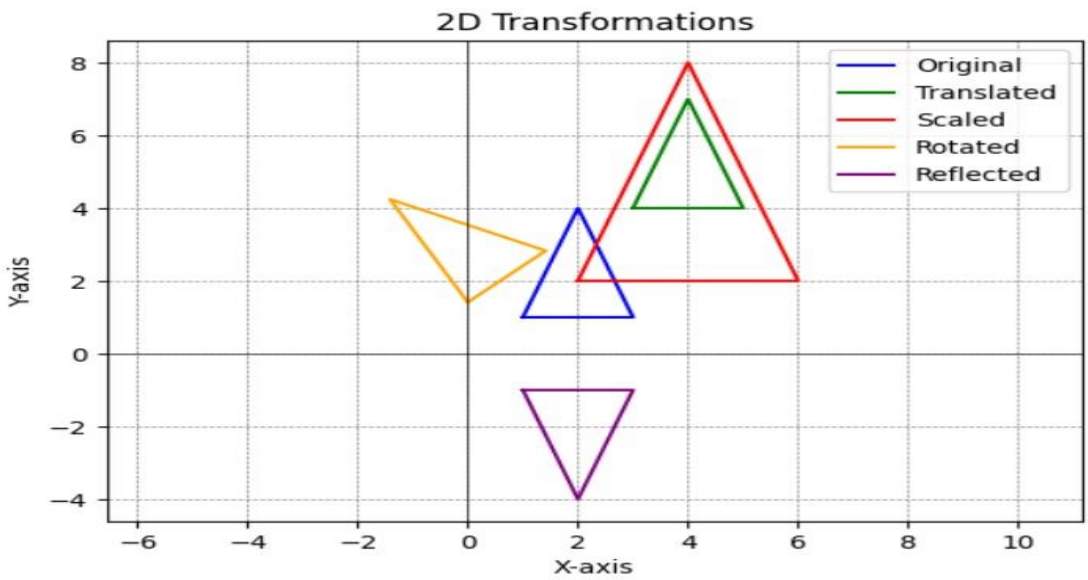
```
plt.ylabel("Y-axis")

plt.axis("equal")

plt.show()
```

Output: -



Conclusion: - Thus, Study and implementation of Two Dimensional Transformations successfully.

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Practical No.-06

Aim: - Study and implementation of Colouring the Pictures.

Theory: -

Colouring pictures is a fundamental task in computer graphics used for visualization, illustration, and animation. It involves filling closed regions with colours, gradients, or patterns.

1. **Colour Representation:**
   - Colours are typically represented in **RGB** (Red, Green, Blue) or **RGBA** (RGB + Alpha for transparency) format.
2. **Filling Shapes:**
   - Closed shapes like polygons, circles, and rectangles can be filled using a fill property or colouring function provided by graphics libraries.
3. **Libraries Used:**
   - **Matplotlib:** Ideal for simple visualizations and supports filling regions with colours.
   - **Pygame:** Provides advanced control for interactive graphics and animations.
4. **Applications:**
   - Visualization, games, animations, and simulations.

Algorithm: -

- Setup**:**
  - Import necessary libraries (e.g., Matplotlib or Pygame).
  - Define the canvas or drawing area.
- Define **Shapes:**
  - Specify the shapes to be drawn (e.g., polygons, circles, rectangles).
- Assign **Colours:**

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- Use RGB or predefined colour codes for filling the shapes.

- Render **Shapes:**
  - Use the library's fill or colour functionality to apply colours.

- Display**:**
  - Render the coloured picture on the screen.

Program Code: -

```
import matplotlib.pyplot as plt


def draw_colored_shapes():

    fig, ax = plt.subplots(figsize=(6, 6))


    # Draw and color a rectangle

    rectangle = plt.Rectangle((1, 1), 3, 2, color='lightblue',
label="Rectangle")

    ax.add_patch(rectangle)


    # Draw and color a circle

    circle = plt.Circle((5, 5), 1.5, color='lightgreen', label="Circle")
```

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```
ax.add_patch(circle)

# Draw and color a triangle
triangle = plt.Polygon([[2, 6], [3, 8], [4, 6]], color='lightcoral',
label="Triangle")
ax.add_patch(triangle)

# Draw and color a polygon
polygon = plt.Polygon([[6, 1], [7, 2.5], [8, 1], [7, 0]],
color='lightgoldenrodyellow', label="Polygon")
ax.add_patch(polygon)

# Configure plot
ax.set_xlim(0, 10)
ax.set_ylim(0, 10)
plt.axhline(0, color='black', linewidth=0.5)
plt.axvline(0, color='black', linewidth=0.5)
ax.set_aspect('equal', adjustable='box')
```

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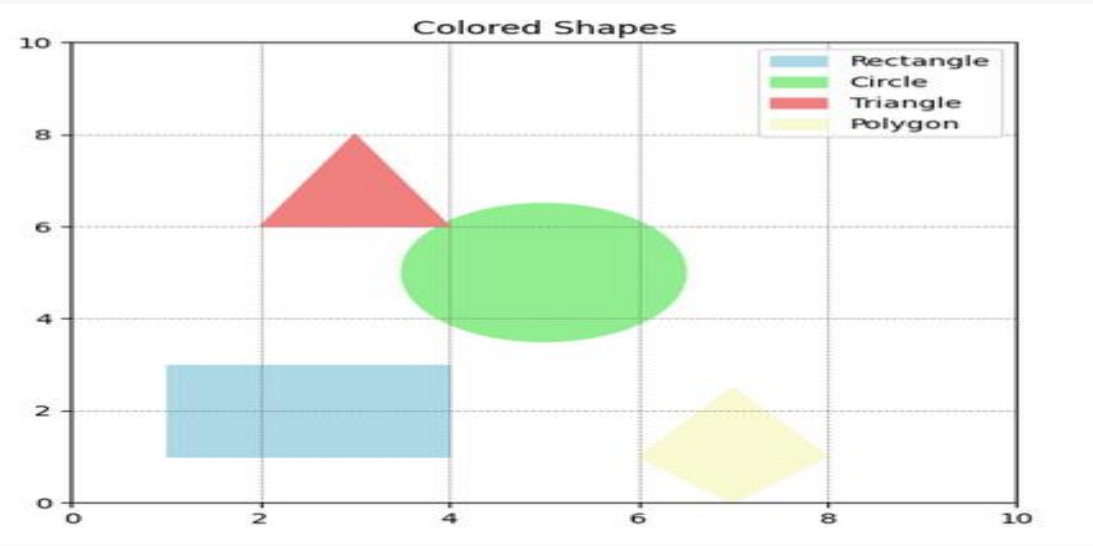
```
plt.grid(color='gray', linestyle='--', linewidth=0.5)

plt.title("Colored Shapes")

plt.legend(loc="best")

plt.show()


# Run the function to display colored shapes

draw_colored_shapes()
```

Output: -



Conclusion: - Thus, Study and implementation of Colouring the Pictures successfully.

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Practical No.-07

Aim: - Create Three Dimensional Transformations.

Theory: -

Three-dimensional transformations are essential operations in computer graphics for manipulating 3D objects. These transformations change an object's position, size, or orientation in a 3D coordinate system.

Types of Transformations:

- Translation**:** Moves an object from one position to another in 3D space.
- Scaling**:** Changes the size of an object in the 3D space.
- Rotation**:** Rotates an object around one of the coordinate axes:
- Reflection**:** Flips an object about a plane.

Algorithm: -

1. **Define the Object:** Represent the 3D object using vertices in a 3D coordinate system.
2. **Transformation Matrices:**
   - Define the transformation matrices for translation, scaling, and rotation.
   - Use matrix multiplication for transformations.
3. **Apply Transformation:** Multiply the transformation matrix with the object's vertex matrix.
4. **Visualization:** Use Python libraries like **Matplotlib** to display the transformed 3D object.

Program Code: -

```
import numpy as np

import matplotlib.pyplot as plt
```

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```
from mpl_toolkits.mplot3d import Axes3D


def plot_3d_object(ax, vertices, edges, color, label):
    for edge in edges:
        x = [vertices[edge[0], 0], vertices[edge[1], 0]]
        y = [vertices[edge[0], 1], vertices[edge[1], 1]]
        z = [vertices[edge[0], 2], vertices[edge[1], 2]]
        ax.plot(x, y, z, color=color, label=label if edge == edges[0] else "")


def translate(vertices, tx, ty, tz):
    translation_matrix = np.array([
        [1, 0, 0, tx],
        [0, 1, 0, ty],
        [0, 0, 1, tz],
        [0, 0, 0, 1]
    ])
    vertices_homogeneous = np.hstack([vertices, np.ones((vertices.shape[0],
1))])
    transformed_vertices = vertices_homogeneous @ translation_matrix.T
    return transformed_vertices[:, :3]
```

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```
def scale(vertices, sx, sy, sz):
    scaling_matrix = np.array([
        [sx, 0, 0, 0],
        [0, sy, 0, 0],
        [0, 0, sz, 0],
        [0, 0, 0, 1]
    ])
    vertices_homogeneous = np.hstack([vertices, np.ones((vertices.shape[0],
1))])
    transformed_vertices = vertices_homogeneous @ scaling_matrix.T
    return transformed_vertices[:, :3]


def rotate_x(vertices, angle):
    rad = np.radians(angle)
    rotation_matrix = np.array([
        [1, 0, 0, 0],
        [0, np.cos(rad), -np.sin(rad), 0],
        [0, np.sin(rad), np.cos(rad), 0],
        [0, 0, 0, 1]
    ])
```

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```
    vertices_homogeneous = np.hstack([vertices, np.ones((vertices.shape[0],
1))])
    transformed_vertices = vertices_homogeneous @ rotation_matrix.T
    return transformed_vertices[:, :3]


def rotate_y(vertices, angle):
    rad = np.radians(angle)
    rotation_matrix = np.array([
        [np.cos(rad), 0, np.sin(rad), 0],
        [0, 1, 0, 0],
        [-np.sin(rad), 0, np.cos(rad), 0],
        [0, 0, 0, 1]
    ])
    vertices_homogeneous = np.hstack([vertices, np.ones((vertices.shape[0],
1))])
    transformed_vertices = vertices_homogeneous @ rotation_matrix.T
    return transformed_vertices[:, :3]


def rotate_z(vertices, angle):
    rad = np.radians(angle)
    rotation_matrix = np.array([
```

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```
        [np.cos(rad), -np.sin(rad), 0, 0],
        [np.sin(rad), np.cos(rad), 0, 0],
        [0, 0, 1, 0],
        [0, 0, 0, 1]
    ])
    vertices_homogeneous = np.hstack([vertices, np.ones((vertices.shape[0],
1))])
    transformed_vertices = vertices_homogeneous @ rotation_matrix.T
    return transformed_vertices[:, :3]

# Example usage
vertices = np.array([[0, 0, 0], [1, 0, 0], [1, 1, 0], [0, 1, 0],  # Bottom square
                     [0, 0, 1], [1, 0, 1], [1, 1, 1], [0, 1, 1]])  # Top square

edges = [(0, 1), (1, 2), (2, 3), (3, 0),  # Bottom square
         (4, 5), (5, 6), (6, 7), (7, 4),  # Top square
         (0, 4), (1, 5), (2, 6), (3, 7)]  # Vertical edges

fig = plt.figure(figsize=(10, 7))
ax = fig.add_subplot(111, projection='3d')
```

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```
# Plot original object
plot_3d_object(ax, vertices, edges, 'blue', 'Original')

# Apply transformations
translated_vertices = translate(vertices, 2, 3, 4)
scaled_vertices = scale(vertices, 1.5, 1.5, 1.5)
rotated_vertices = rotate_y(vertices, 45)

# Plot transformed objects
plot_3d_object(ax, translated_vertices, edges, 'green', 'Translated')
plot_3d_object(ax, scaled_vertices, edges, 'red', 'Scaled')
plot_3d_object(ax, rotated_vertices, edges, 'orange', 'Rotated')

# Configure the 3D plot
ax.set_title("3D Transformations")
ax.set_xlabel("X-axis")
ax.set_ylabel("Y-axis")
ax.set_zlabel("Z-axis")
plt.legend()
plt.show()
```

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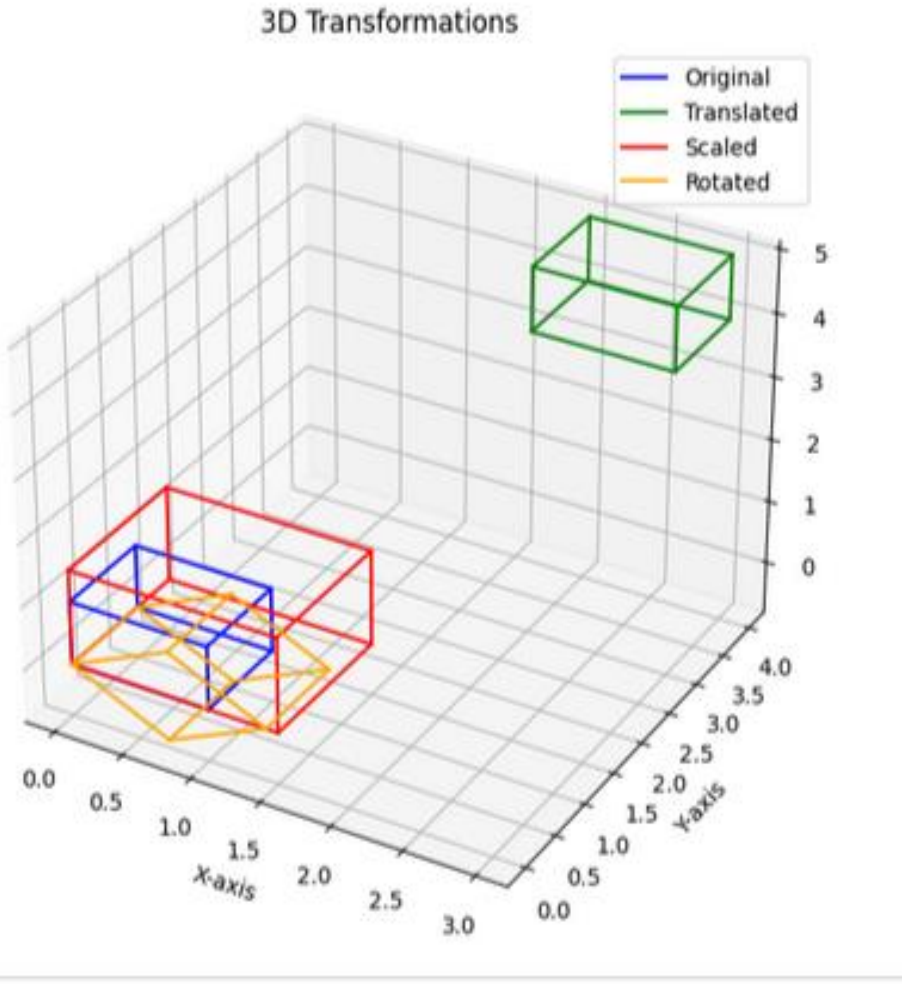
Output: -



Conclusion: - Thus, Create Three Dimensional Transformations successfully.

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Practical No.-08

Aim: - Implementation of Curve Generation.

Theory: -

Curve generation is a fundamental aspect of computer graphics, used for rendering smooth and continuous shapes. Curves are defined mathematically and can be represented by equations or control points.

Types of Curves

1. **Bezier Curves:**
   - Defined using control points.
   - The curve lies entirely within the convex hull of its control points.
2. **B-Splines:**
   - Generalization of Bezier curves.
   - Combines multiple Bezier segments for higher flexibility.
3. **Parametric Curves:**
   - Represented by equations in terms of a parameter

Applications

- Used in animations, CAD systems, and graphical user interfaces.

Algorithm: -

1. **Define Control Points:** Specify the points that determine the shape of the curve.
2. **Select Curve Type:** Choose a curve type such as Bezier or parametric.
3. **Generate Points:** Compute intermediate points on the curve using the curve equations.
4. **Plot the Curve:** Use Python libraries like **Matplotlib** to render the curve.

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Program Code: -

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.special import comb


def bezier_curve(control_points, num_points=100):
    """
    Generate a Bézier curve using control points.
    :param control_points: List of control points (x, y).
    :param num_points: Number of points on the curve.
    :return: Curve points as a numpy array.
    """
    n = len(control_points) - 1  # Degree of the curve
    t = np.linspace(0, 1, num_points)
    curve = np.zeros((num_points, 2))

    for i in range(n + 1):
        binomial_coeff = comb(n, i)
        term = binomial_coeff * ((1 - t) ** (n - i)) * (t ** i)
```

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```
        curve += np.outer(term, control_points[i])

    return curve

def plot_curve(control_points, curve_points):
    """
    Plot the Bézier curve and its control points.
    :param control_points: Control points (x, y).
    :param curve_points: Points on the curve.
    """
    control_points = np.array(control_points)
    plt.figure(figsize=(8, 6))

    # Plot control points and polygon
    plt.plot(control_points[:, 0], control_points[:, 1], 'o--', label='Control Polygon', color='gray')

    # Plot Bézier curve
    plt.plot(curve_points[:, 0], curve_points[:, 1], label='Bézier Curve', color='blue')
```

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```
plt.title('Bézier Curve')
plt.xlabel('X')
plt.ylabel('Y')
plt.legend()
plt.grid(True)
plt.show()

# Define control points
control_points = [(0, 0), (1, 2), (3, 3), (4, 0)]

# Generate Bézier curve
curve_points = bezier_curve(control_points)

# Plot the curve
plot_curve(control_points, curve_points)
```

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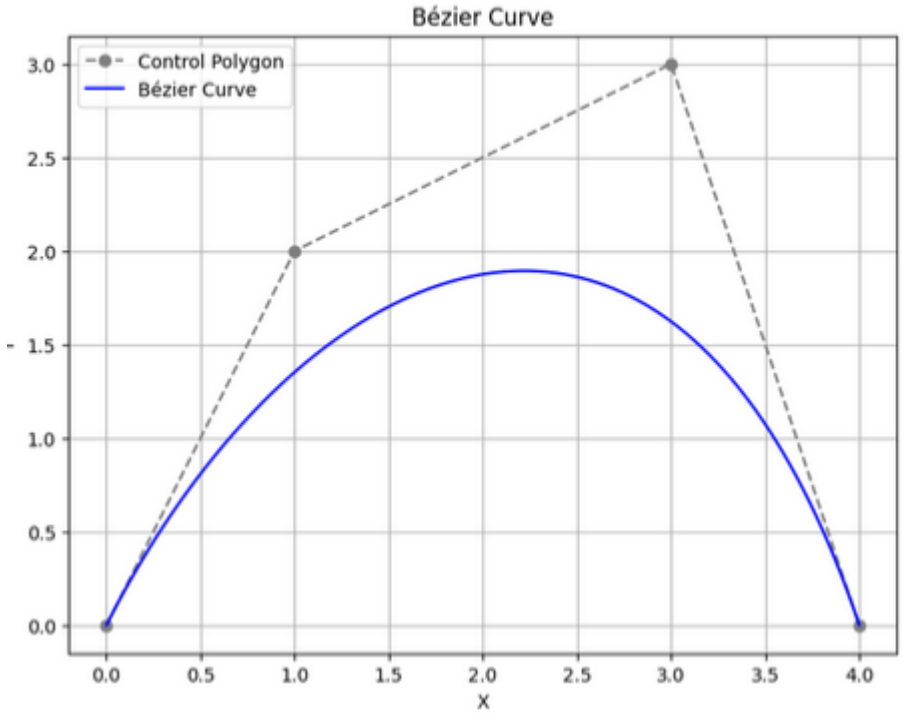
Output: -



Conclusion: - Thus, Implementation of Curve Generation successfully.

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